

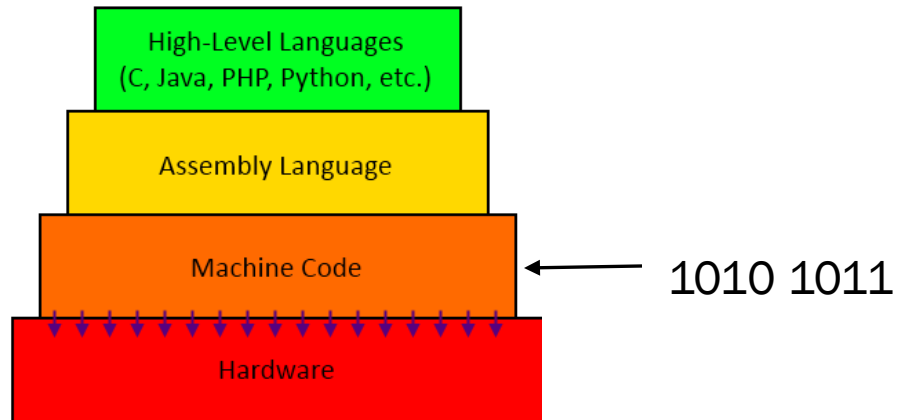
Programming Tools

AN INTRODUCTION



What is a Programming Language?

A programming language is a computer language that is used by programmers (developers) to communicate with computers. It is a set of instructions written in any specific language (C, C++, Java, Python) to perform a specific task.



Programming Languages

- Rust (2010)
- Ruby, PHP, JavaScript (1995)
- Python (1991)
- C++ (1983)
- Since their creation, most languages are regularly updated. For example, C++ version 20 and Python version 3.8.5 both released in 2020.

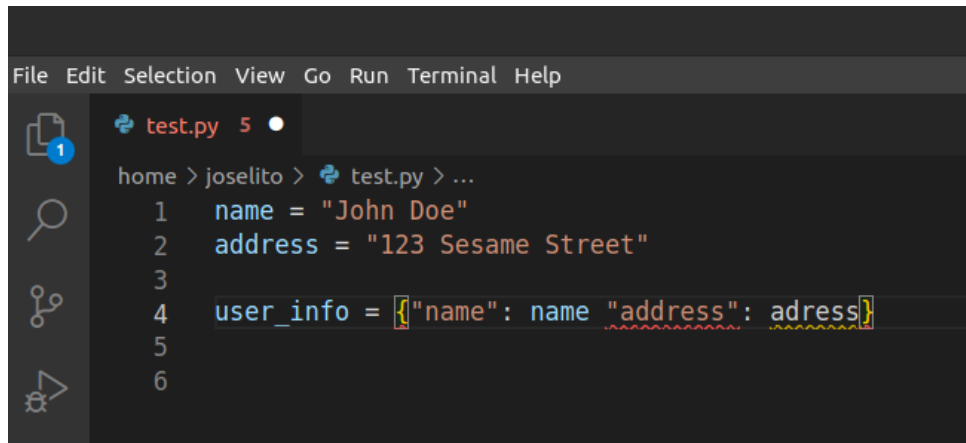


Programming Tools

- Text Editors
- Compilers
- Version Control Systems
- IDEs

Text Editor

A text editor is program that allows you to open, view, and edit plain text files. Software programmers use text editors to write and edit in programming languages. This is one of the primary purposes of text editors, and many of the features of text editing software are built to help these users read and write code.

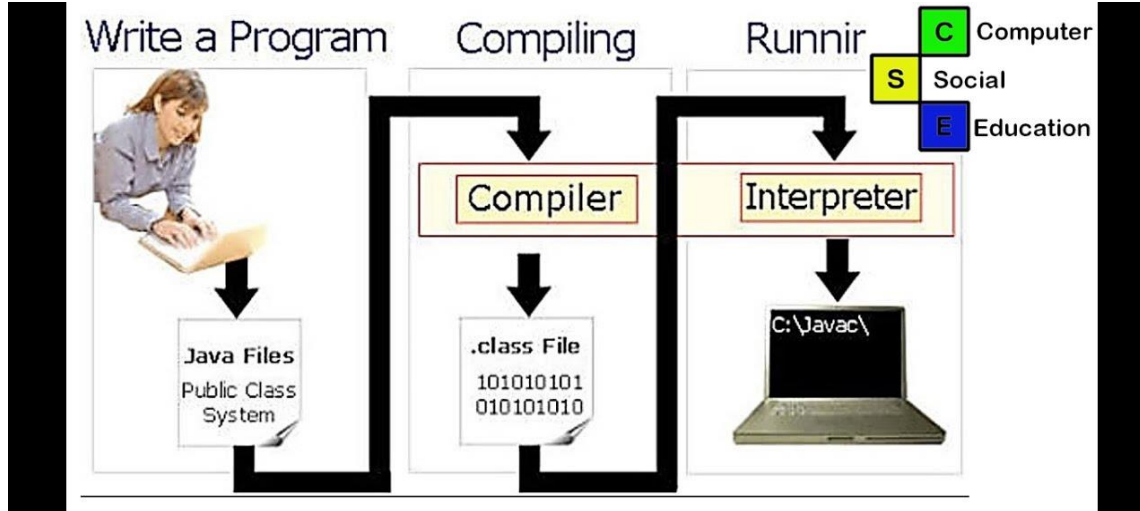


The screenshot shows a text editor window with a dark theme. The menu bar at the top includes 'File', 'Edit', 'Selection', 'View', 'Go', 'Run', 'Terminal', and 'Help'. Below the menu bar, there is a tab labeled 'test.py 5'. The main editing area shows a file path 'home > joselito > test.py > ...' followed by six lines of Python code. The code defines variables for 'name' and 'address', and then creates a dictionary 'user_info' using these variables. The code is as follows:

```
1 name = "John Doe"
2 address = "123 Sesame Street"
3
4 user_info = {"name": name, "address": address}
5
6
```

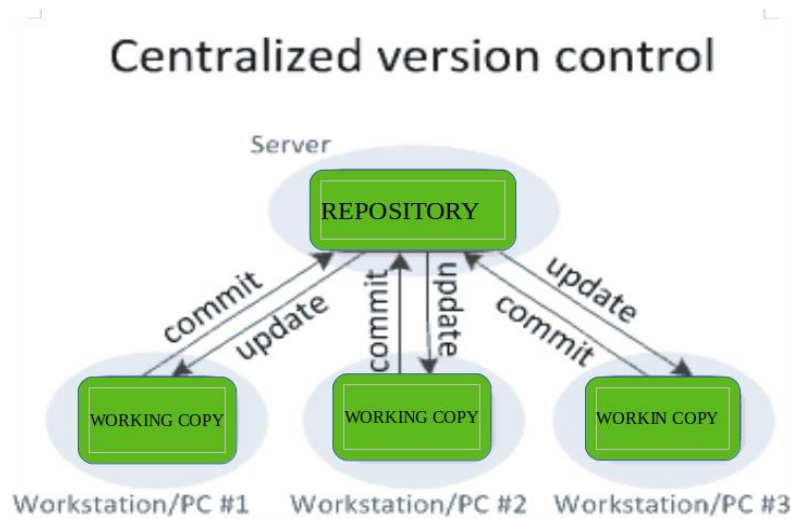
Compiler

A compiler is a special program that translates a programming language's source code into machine code. The outputted machine code is made up entirely of binary bits – 1s and 0s – so it can be read and executed by the processors on the target computers.



Version Control System

A version control system is a kind of software that helps the developer team to efficiently communicate and manage(track) all the changes that have been made to the source code along with the information like who made and what changes have been made.



Integrated Development Environment (IDE)

An integrated development environment (IDE) is software for building applications that combines common developer tools into a single graphical user interface (GUI). An IDE has a text editor, debugger, file manager, compiler, interpreter, VCS all in one.

Examples

- Visual Studio
- Replit
- PyCharm
- Android Studio

